

Mike Hall

Founding Staff Engineer (0-to-1 Product & AI Systems)

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Hands-on Founding Staff Engineer who translates product ambiguity into robust, production-ready software. Combines full-stack web development (Rails, PostgreSQL, modern JS/TypeScript) with practical local-first AI engineering (pgvector embeddings, local LLM orchestration, prompt injection defense, and browser automation).

EXPERIENCE & LEADERSHIP

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Target Focus: Built practical AI prototypes with a hackathon team, introduced local-first workflows through Geekfest@OMF, and helped establish safer ways to apply emerging tools to platform work.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.
- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Principal Architect

Phalanx Duel | Independent Project | January 2022 - Present

Target Focus: Built a server-authoritative deterministic game platform in TypeScript and Node.js: pure rules engine, append-only action ledger, WebSocket real-time sync, and property-based Truth Gate verification gates across the full state lifecycle.

- Designed a deterministic engine with replayable action logs and server-authoritative state transitions, making complex outcomes reproducible and independently verifiable.
- Built replay validation, adversarial coverage, and CI verification gates so state transitions can be checked across the complete lifecycle.
- Uses controlled agent workflows with bounded tasks, acceptance criteria, automated checks, and human-reviewed outputs rather than treating model output as authority.

Principal Architect

WWWorkRemote | Independent Project | January 2021 - Present

Target Focus: Architected multi-source Rails ingestion pipeline, pgvector semantic search, local-first YAML model registry, 4-stage prompt injection defense, and 16-adapter MV3 Chrome extension.

- **Multi-Source Data Ingestion:** Built a multi-source Rails ingestion pipeline and PostgreSQL storage schema, unifying fragmented job market datasets for structured downstream analysis.
- **Semantic Search & Embeddings:** Implemented vector search embeddings via pgvector, enabling high-precision semantic matching between complex candidate profiles and job requirements.
- **Local-First AI Orchestration:** Designed provider-agnostic LLM orchestration layer (llama.cpp/Ollama with hosted Claude fallback), enforcing local-first inference for sensitive data with per-call model routing.
- **Prompt Injection Defense:** Built a 4-stage guardrails pipeline screening untrusted scraped text via weighted heuristics and instruction-density scoring before model ingestion.
- **Browser Extension & Lifecycle Capture:** Built an MV3 Chrome extension with 16 provider adapters that extracts postings from live ATS pages and captures full application lifecycles.
- **Architectural Constraint Injection:** Injected static analysis rules and architectural constraints into model context windows, preventing drift between generated outputs and security contracts.

Principal Engineer & Curator

Technical Conversation Archive | Independent Project | January 2011 - Present

Target Focus: Built local-only Whisper transcription pipeline, semantic taxonomy cross-linking, and build-time schema enforcement gates for a multi-decade tech interview media corpus.

- **Local-Only AI Restoration Pipeline:** Built a Whisper and local-LLM pipeline that transcribes, diarizes, and structures long-form interview audio entirely on-device, keeping a large private media corpus off third-party inference services.
- **Schema-Validated Content Platform:** Runs on a contract-validated data layer: every interview, asset, and transcript is checked against an explicit schema at build time, with referential-integrity checks that fail the build rather than publishing broken data.
- **Semantic Cross-Linking & Taxonomy Generation:** Generates a topic taxonomy, knowledge graph, and semantic cross-links across the corpus, turning an unstructured media archive into navigable, individually indexed pages.
- **Primary-Source Capture:** Conducted the original interviews on-site at GOTO Conference, Software Craftsmanship North America, RailsConf, and WindyCityRails, with practitioners including Dave Thomas, Stuart Halloway, Corey Haines, Sandro Mancuso, and Micah Martin.
- **Digital Archaeology:** Recovered and reconstructed material from defunct platforms and web archives, reconciling incomplete metadata across sources to restore provenance for recordings that would otherwise have been lost.

SELECTED PRODUCTION PROJECTS

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | January 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- **Bidirectional Knowledge Server:** Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.

- **Telemetry-Driven Diagnosis:** Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- **Context as a Budgeted Resource:** Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

CORE SKILLS & COMPETENCIES

01 • 0-to-1 Full-Stack Architecture & Rapid Prototyping	02 • Local-First AI Orchestration (llama.cpp, whisper.cpp, Claude API)	03 • Vector Embeddings & Semantic Search (PostgreSQL pgvector)	04 • Prompt Injection Defense & Multi-Stage Guardrails
05 • AI Agent Workflows & MCP Server Integration (Python, Node.js)	06 • Automated Schema Inference & Dynamic JSON Pipelines	07 • Ruby on Rails, Node.js, WebSockets & Real-Time Sync	08 • High Agency, Autonomous Problem Solving

ADDITIONAL EXPERIENCE

- **Open-Source Transition Lead, Coderwall** | January 2014 - December 2014 | Remote: Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Senior Software Developer, ReachLocal** | March 2015 - November 2016 | Remote: Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- **Senior Backend Developer, SK Holdings, Inc.** | January 2019 - December 2020 | Chicago, IL: Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.
- **Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems), Groupon** | July 2011 - May 2013 | Chicago, IL: Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.
- **Senior .NET Developer, TicketsNow** | November 2005 - March 2007 | Crystal Lake, IL: Owned real-time inventory systems, leading transactional integrity and iterative delivery for revenue-critical operations.

EARLIER EXPERIENCE

Selected earlier experience

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that inform my leadership today.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.